Evaluating and Graphing Exponential Functions

Evaluating exponential rules, building tables and graphs, and solving equations with matching powers

Directions

- In $a \cdot b^x$ the exponent belongs to b alone. Work out the power first, then multiply by a.
- $b^0 = 1$ for every base, and $b^{-n} = \dfrac{1}{b^n}$ — a negative exponent gives a small positive value, never a negative one.
- To solve $a \cdot b^x = c$, divide by a, then write the right-hand side as a power of b and match exponents.
- Plot each table point carefully and join the points with a smooth curve, not straight segments.

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1. Let $f(x) = 3 \cdot 2^x$.

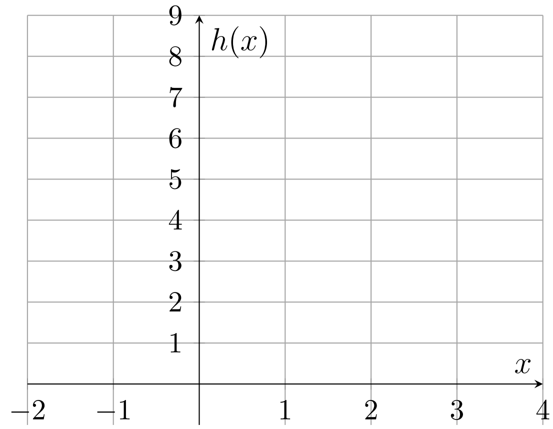
(a) $f(0) =$ _________ (b) $f(3) =$ _________

2. Let $g(x) = 5^x$.

(a) $g(2) =$ _________ (b) $g(0) =$ _________

3. Let $h(x) = 2^x$. Complete the table, then plot the five points and join them with a smooth curve.

x	-1	0	1	2	3
$h(x)$	________	________	________	________	________



4. Let $p(x) = 100 \cdot (0.5)^x$.

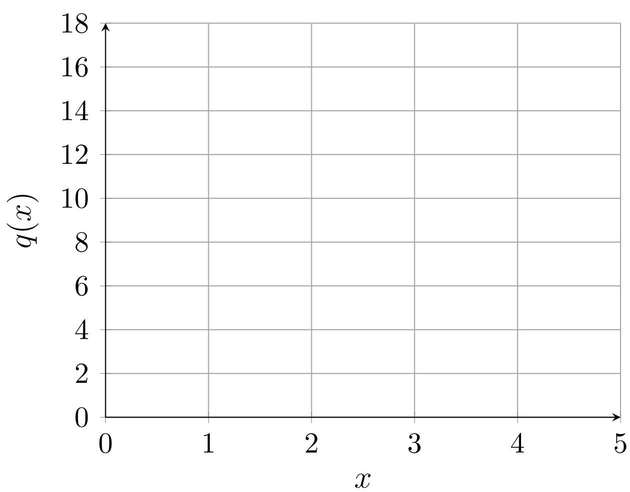
(a) $p(0) =$ ________ (b) $p(3) =$ ________

5. Solve $2^x = 64$. Write 64 as a power of 2 first.

Answer: ____________________

6. Let $q(x) = 16 \cdot \left(\frac{1}{2}\right)^x$. Complete the table, then plot the points and join them with a smooth curve.

x	0	1	2	3	4
$q(x)$	________	________	________	________	________



7. An account of \$800 grows by 5% a year, so its balance after t years is $800 \cdot 1.05^t$ dollars. Find the balance after 6 years, to two decimal places.

Answer: ________________

8. Solve $5 \cdot 3^x = 405$.

Answer: ________________

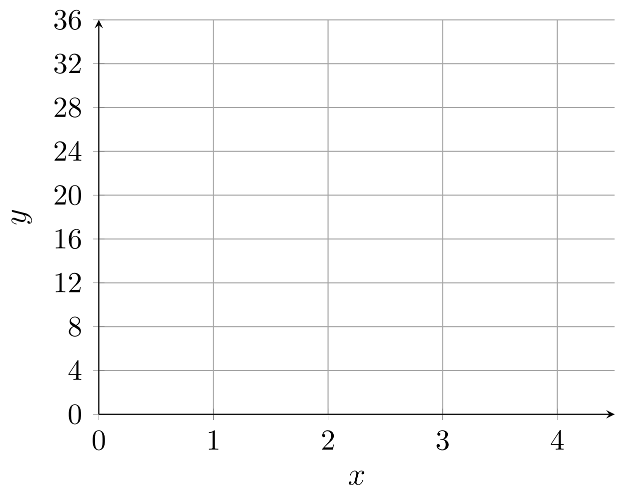
9. Let $f(x) = 2 \cdot 3^x$.

(a) $f(4) =$ ________

(b) $f(-2) =$ ________ (give an exact fraction)

10. Let $y = 4 \cdot 2^x$. Complete the table, plot the points, and join them with a smooth curve.

x	0	1	2	3
y	________	________	________	________



Solve $4 \cdot 2^x = 64$ to find where the curve reaches 64. $x =$ ________

Explain why this curve never touches the horizontal axis, no matter how far left it is extended.

11. Solve $\left(\frac{1}{2}\right)^x = \frac{1}{32}$.

Answer: ________________

12. A bacterial culture doubles every 3 hours. Starting from 500 cells, the count after t hours is $500 \cdot 2^{t/3}$.

(a) How many cells after 9 hours? ________

(b) How many cells after 5 hours, to two decimal places? ________

(c) After how many hours does the count reach 16,000? $t =$ ________